

Figure 13.6 Entity type ORDER showing repeating groups of attributes

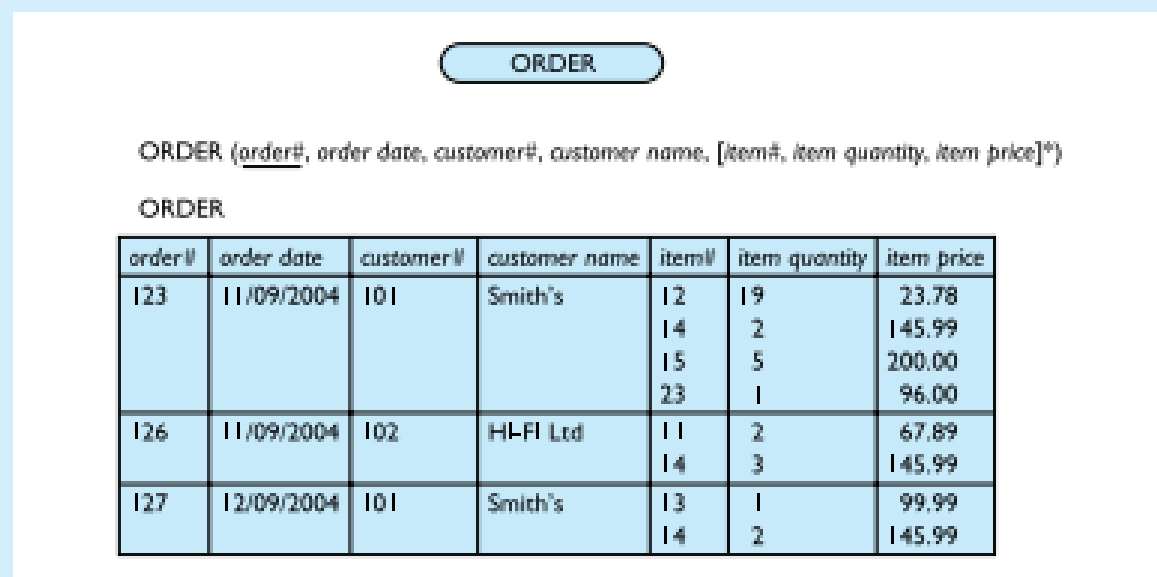
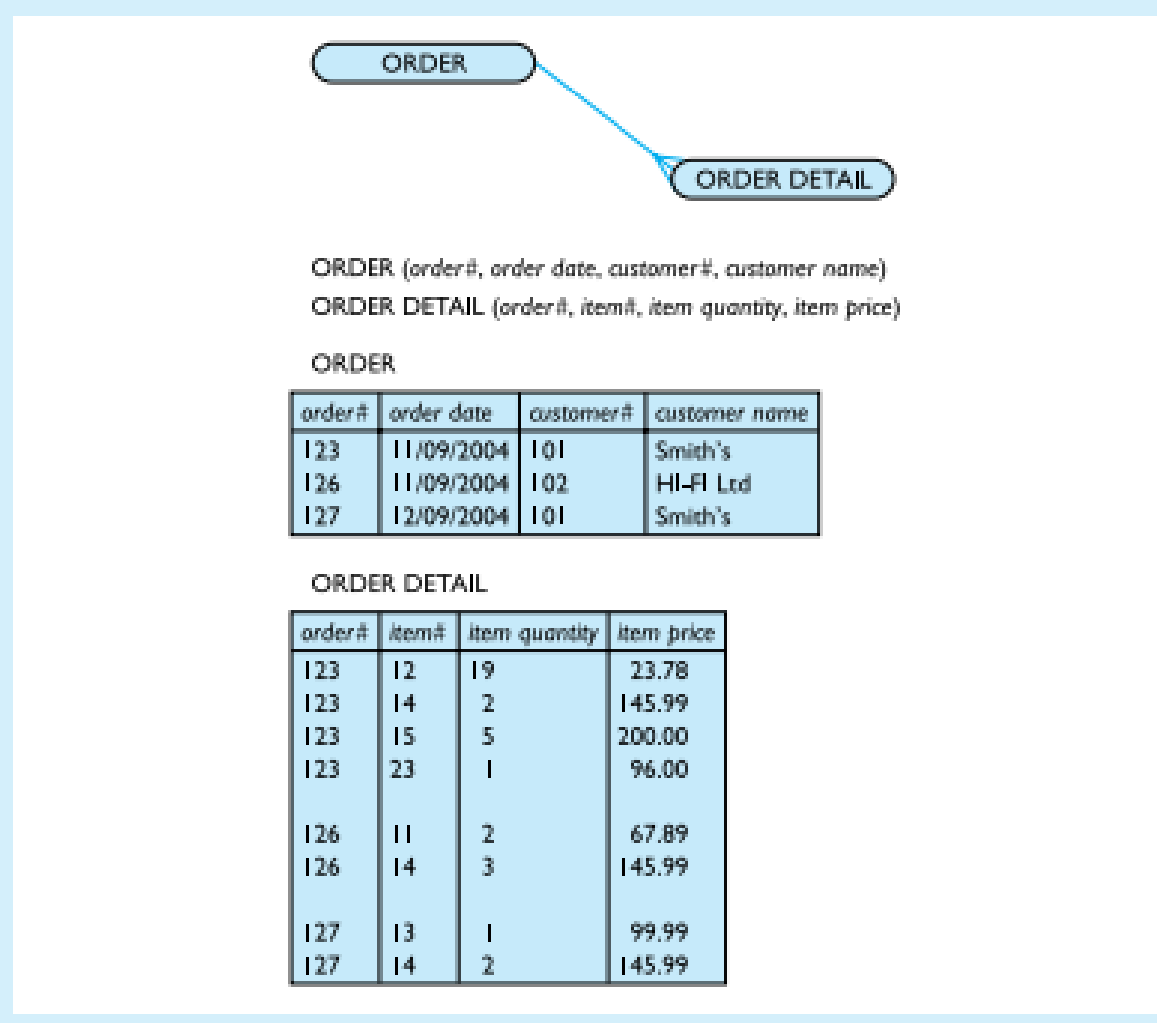


Figure 13.7 Entities in first normal form



order where the non-repeating attributes, such as *order date* and *customer#*, appear. The key for ORDER DETAIL is the set of attributes consisting of *order#*, *item#*. Each on its own will not identify a unique occurrence of ORDER DETAIL, but together they guarantee this.

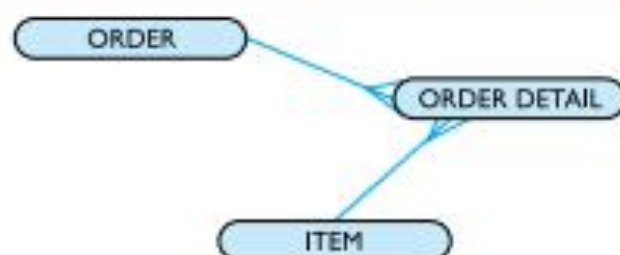
Second normal form

An entity is in second normal form if it is in 1NF and every attribute that is not part of the key depends on the whole key.

In Figure 13.7, the entity type ORDER has only one attribute as its key, so all the other attributes depend on this attribute. Considering ORDER DETAIL, *item quantity* depends on the whole key. This is because the quantity of an item in an order does not depend only on the item (the item is ordered in other orders) or only on the order (there are other items ordered in the order). However, the combination of *order#* and *item#* fixes *item quantity*. The same is not true of *item price*. This depends only on *item#*. When it is realized that order details are about items ordered, it becomes clear that ITEM is an entity type.

This is shown in Figure 13.8. Each order detail is about only one item, but each item may be referred to by many examples of order detail from different orders. The relationship is 1:n as shown. The advantage of second normal form is that data on an

Figure 13.8 Entities in second normal form



ORDER (order#, order date, customer#, customer name)

ORDER DETAIL (order#, item#, item quantity)

ITEM (item#, item price)

ORDER

order#	order date	customer#	customer name
123	11/09/2004	101	Smith's
126	11/09/2004	102	Hi-Fi Ltd
127	12/09/2004	101	Smith's

ORDER DETAIL

order#	item#	item quantity
123	12	19
123	14	2
123	15	5
123	23	1
126	11	2
126	14	3
127	13	1
127	14	2

ITEM

item#	item price
11	67.89
12	23.78
13	99.99
14	145.99
15	200.00
23	96.00

item such as its price occurs only once. This contrasts with the repeated prices for items with *item#* 1 and 2 in Figure 13.7. The data model will be used to design an effective database. A good database design minimizes the duplication of data.

Third normal form

An entity is in third normal form if it is in 2NF and each attribute (or set of attributes) that is not part of the key depends on nothing but the key.

In Figure 13.8, both ORDER DETAIL and ITEM contain only one non-key attribute each. So each of these attributes can depend, by definition, only on the key. However, the entity type ORDER has an attribute *customer name* that depends on the *customer #*. Customer name is a fact about the customer. Customers place orders.

This is shown in Figure 13.9. A new entity type CUSTOMER is introduced. This has a key, *customer#*, and one non-key attribute, *customer name*. Each customer may place

Figure 13.9 Entities in third normal form

